EDA for the road features extracted from images and contextual database

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2026-09-09

EDA Report: Dataset (model_dataset 1.xlsx)

Introduction

This report presents an exploratory data analysis (EDA) of `model_dataset 1.xlsx`, a dataset of 49 road segments in Western Australia. The contextual variables (e.g. population statistics, distance to public transport, health, and education facilities, building and retail counts) were extracted from a contextual ecosocial database, while the image-derived variables (e.g. footpath type, land use, and the presence of nearby amenities) were obtained from earlier feature extraction work. The goal is to identify variables that are associated with observed pedestrian counts, in preparation for building a predictive regression model.

Dataset and Variables

The dataset contains two response variables, measuring pedestrian counts per hour in two directions:

- `obs_peak_along_ped_per_h`
- `obs_peak_cross_ped_per_h`

The remaining variables fall into two groups: 21 numeric variables (e.g. building counts, population density, distance to public transport, health, and education facilities) and 16 categorical variables (e.g. footpath type, land use, and the presence of nearby amenities).

```
## tibble [49 x 39] (S3: tbl_df/tbl/data.frame)
##  $ obs_peak_cross_ped_per_h                      : num [1:49] 0.333 4 47 16 57.333 ...
##  $ obs_peak_along_ped_per_h                      : num [1:49] 12.33 7.67 101.33 11 108 ...
##  $ abs_mb_area_sqkm_2021                         : num [1:49] 7.9418 0.3167 0.1362 0.0568 0.0616 .
##  $ abs_mb_dwellings_2021                         : num [1:49] 35 0 0 0 0 36 0 12 0 0 ...
##  $ abs_mb_persons_usual_resident_2021            : num [1:49] 74 0 0 0 0 83 0 27 0 0 ...
##  $ abs_mb_dwellings_per_sqkm_2021                : num [1:49] 4.41 0 0 0 0 ...
##  $ abs_mb_persons_per_sqkm_2021                  : num [1:49] 9.32 0 0 0 0 ...
##  $ pta_nearest_active_stop_distance_m            : num [1:49] 9163.3 564.3 265 93.5 182.2 ...
##  $ pta_active_stops_within_500m                  : num [1:49] 0 0 4 11 13 0 1 2 1 4 ...
##  $ health_nearest_establishment_distance_m       : num [1:49] 29477 2886 1237 486 1156 ...
##  $ education_nearest_active_school_distance_m    : num [1:49] 20481 1350 319 726 238 ...
##  $ ms_building_count_500m                        : num [1:49] 12 175 370 392 329 222 286 340 287 3
##  $ ms_building_count_per_sqkm_500m               : num [1:49] 15.3 222.9 471.3 499.3 419.1 ...
##  $ ms_building_footprint_area_m2_500m            : num [1:49] 2321 122504 136178 98962 103976 ...
##  $ ms_building_coverage_ratio_500m               : num [1:49] 0.00296 0.15604 0.17346 0.12605 0.13
##  $ overture_food_drink_count_500m                : num [1:49] 0 3 19 14 20 0 14 5 5 14 ...
##  $ overture_grocery_convenience_pharmacy_count_500m: num [1:49] 0 0 8 13 7 0 5 4 5 8 ...
##  $ overture_other_shopping_count_500m            : num [1:49] 0 19 35 20 19 0 11 6 8 30 ...
##  $ pedestrian_crossings                          : num [1:49] 0 0 1 0 1 1 1 0 0 0 ...
```

```
##  $ commercial_buildings                              : num [1:49] 0 0 6 1 8 1 8 6 4 12 ...
##  $ residential_buildings                             : num [1:49] 0 0 0 7 2 5 0 0 0 2 ...
##  $ industrial_buildings                              : num [1:49] 0 3 0 0 0 0 2 0 0 0 ...
##  $ parking_areas                                     : num [1:49] 0 1 3 2 4 2 2 3 3 3 ...
##  $ abs_mb_category_2021                              : chr [1:49] "Residential" "Industrial" "Transpor
##  $ pta_nearest_active_stop_type                      : chr [1:49] "Transwa" "Transwa" "School" "School
##  $ health_nearest_establishment_category             : chr [1:49] "Acute Hospital" "Acute Hospital" "A
##  $ health_nearest_establishment_type                 : chr [1:49] "Public" "Public" "Public" "Public"
##  $ education_nearest_school_sector                   : chr [1:49] "NON-GOVERNMENT" "GOVERNMENT" "GOVERN
##  $ education_nearest_school_low_year                 : chr [1:49] "KIN" "KIN" "Y07" "KIN" ...
##  $ education_nearest_school_high_year                : chr [1:49] "Y06" "Y06" "Y12" "Y06" ...
##  $ overture_shopping_mall_present_500m               : logi [1:49] FALSE FALSE TRUE FALSE TRUE FALSE .
##  $ footpath_both_sides                               : logi [1:49] FALSE TRUE TRUE FALSE TRUE FALSE ..
##  $ footpath_one_side                                 : logi [1:49] FALSE FALSE TRUE TRUE TRUE TRUE ...
##  $ fencing_barriers                                  : logi [1:49] TRUE TRUE TRUE TRUE TRUE TRUE ...
##  $ railway_station                                   : logi [1:49] FALSE FALSE TRUE FALSE FALSE FALSE
##  $ bus_stops                                         : logi [1:49] FALSE FALSE FALSE FALSE FALSE FALSE
##  $ overpass_bridge                                   : logi [1:49] FALSE FALSE FALSE FALSE FALSE FALSE
##  $ park_bushland_reserve                             : logi [1:49] TRUE TRUE TRUE FALSE TRUE TRUE ...
##  $ shared_pedestrian_and_cycle_path                  : logi [1:49] FALSE TRUE TRUE FALSE FALSE TRUE ..
```

Numeric Variables

Correlation coefficients were calculated between each numeric variable and both response variables.

```
##                                                obs_peak_cross_ped_per_h
## abs_mb_area_sqkm_2021                                             -0.15
## abs_mb_dwellings_2021                                             -0.05
## abs_mb_persons_usual_resident_2021                                 0.05
## abs_mb_dwellings_per_sqkm_2021                                    -0.12
## abs_mb_persons_per_sqkm_2021                                      -0.01
## pta_nearest_active_stop_distance_m                                 0.16
## pta_active_stops_within_500m                                      -0.06
## health_nearest_establishment_distance_m                            0.00
## education_nearest_active_school_distance_m                        -0.15
## ms_building_count_500m                                             0.08
## ms_building_count_per_sqkm_500m                                    0.08
## ms_building_footprint_area_m2_500m                                 0.10
## ms_building_coverage_ratio_500m                                    0.10
## overture_food_drink_count_500m                                     0.57
## overture_grocery_convenience_pharmacy_count_500m                   0.39
## overture_other_shopping_count_500m                                 0.29
## pedestrian_crossings                                               0.23
## commercial_buildings                                               0.66
## residential_buildings                                             -0.21
## industrial_buildings                                              -0.08
## parking_areas                                                      0.42
##                                               obs_peak_along_ped_per_h
## abs_mb_area_sqkm_2021                                             -0.13
## abs_mb_dwellings_2021                                             -0.07
## abs_mb_persons_usual_resident_2021                                -0.03
## abs_mb_dwellings_per_sqkm_2021                                    -0.09
## abs_mb_persons_per_sqkm_2021                                      -0.05
## pta_nearest_active_stop_distance_m                                -0.08
## pta_active_stops_within_500m                                       0.02
```

```
## health_nearest_establishment_distance_m                    -0.01
## education_nearest_active_school_distance_m                 -0.10
## ms_building_count_500m                                      0.09
## ms_building_count_per_sqkm_500m                             0.09
## ms_building_footprint_area_m2_500m                          0.12
## ms_building_coverage_ratio_500m                             0.12
## overture_food_drink_count_500m                              0.59
## overture_grocery_convenience_pharmacy_count_500m            0.42
## overture_other_shopping_count_500m                          0.42
## pedestrian_crossings                                        0.18
## commercial_buildings                                        0.77
## residential_buildings                                      -0.14
## industrial_buildings                                       -0.08
## parking_areas                                               0.46
```

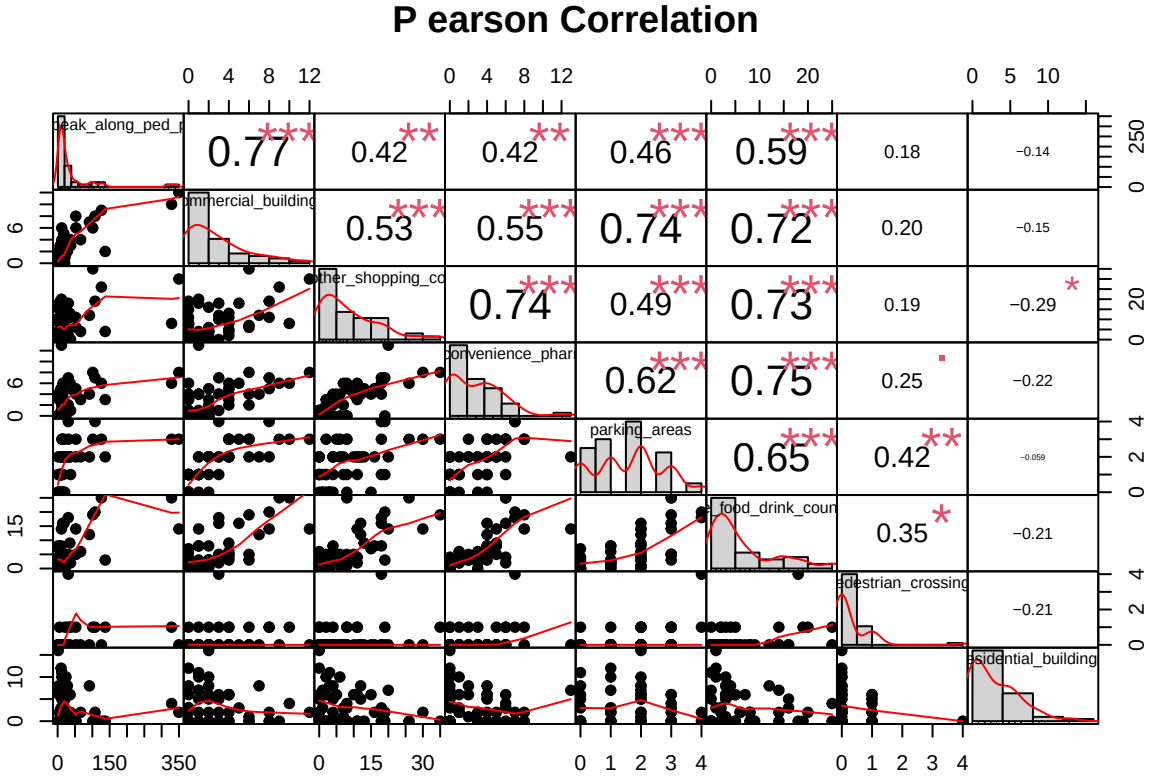
Findings:

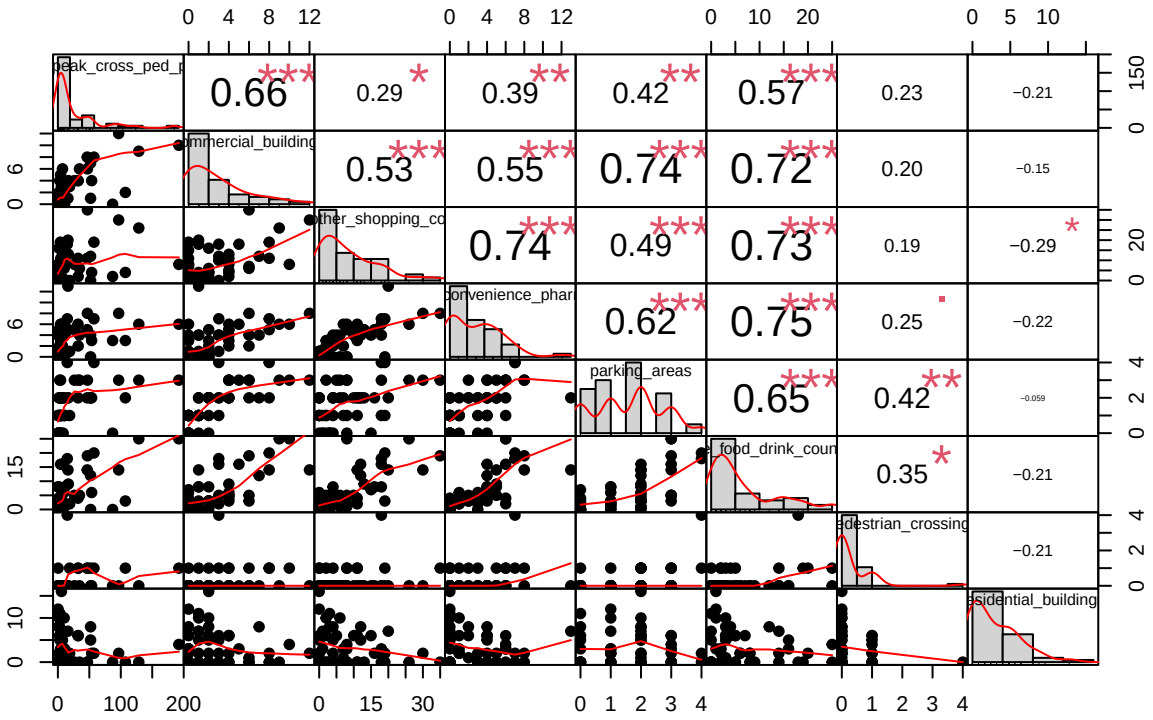
Three variables show a clear correlation (|r| >= 0.4) with both response variables: `commercial_buildings` (r = 0.66–0.77), `overture_food_drink_count_500m` (r = 0.57–0.59), and `parking_areas` (r = 0.42–0.46). All three are statistically significant ($p < 0.01$) and relate to commercial and retail activity near the site.

Two further retail-related variables, `overture_grocery_convenience_pharmacy_count_500m` and `overture_other_shopping_count_500m`, show a moderate correlation with `obs_peak_along_ped_per_h` (r = 0.42 for both) but a weaker correlation with `obs_peak_cross_ped_per_h` (r = 0.39 and 0.29 respectively).

`pedestrian_crossings` and `residential_buildings` show weak correlations (|r| between 0.14 and 0.23) that are not statistically significant.

The remaining variables, including ABS population statistics, distance to public transport, health, and education facilities, and Microsoft building footprint measures, show little to no correlation (|r| < 0.2) with either target.





The five significant predictors are also highly correlated with each other (r = 0.56 to 0.87). This indicates strong multicollinearity, which will need to be addressed before these variables can be used together in a regression model.

Logical Categorical Variables

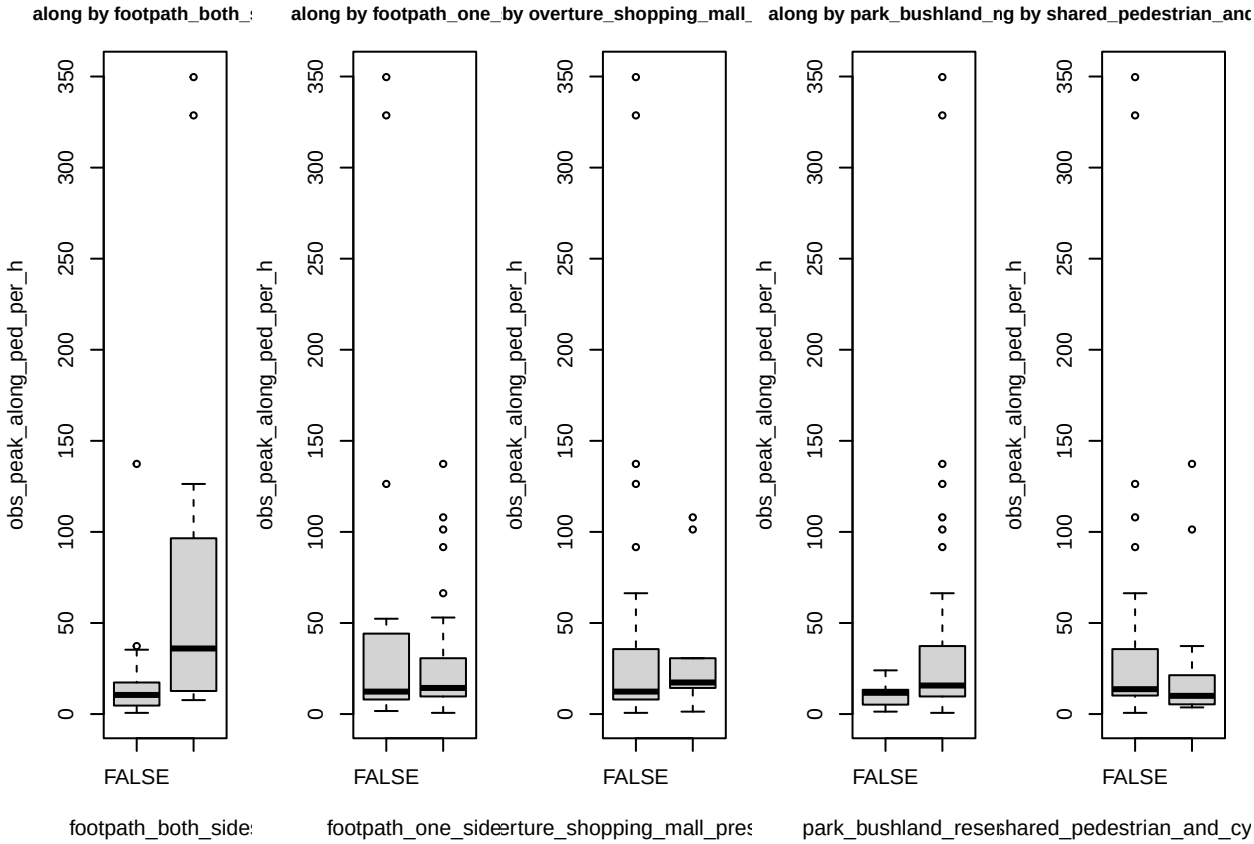
The number of TRUE and FALSE cases was checked for each logical variable, to identify variables with insufficient sample size in either group.

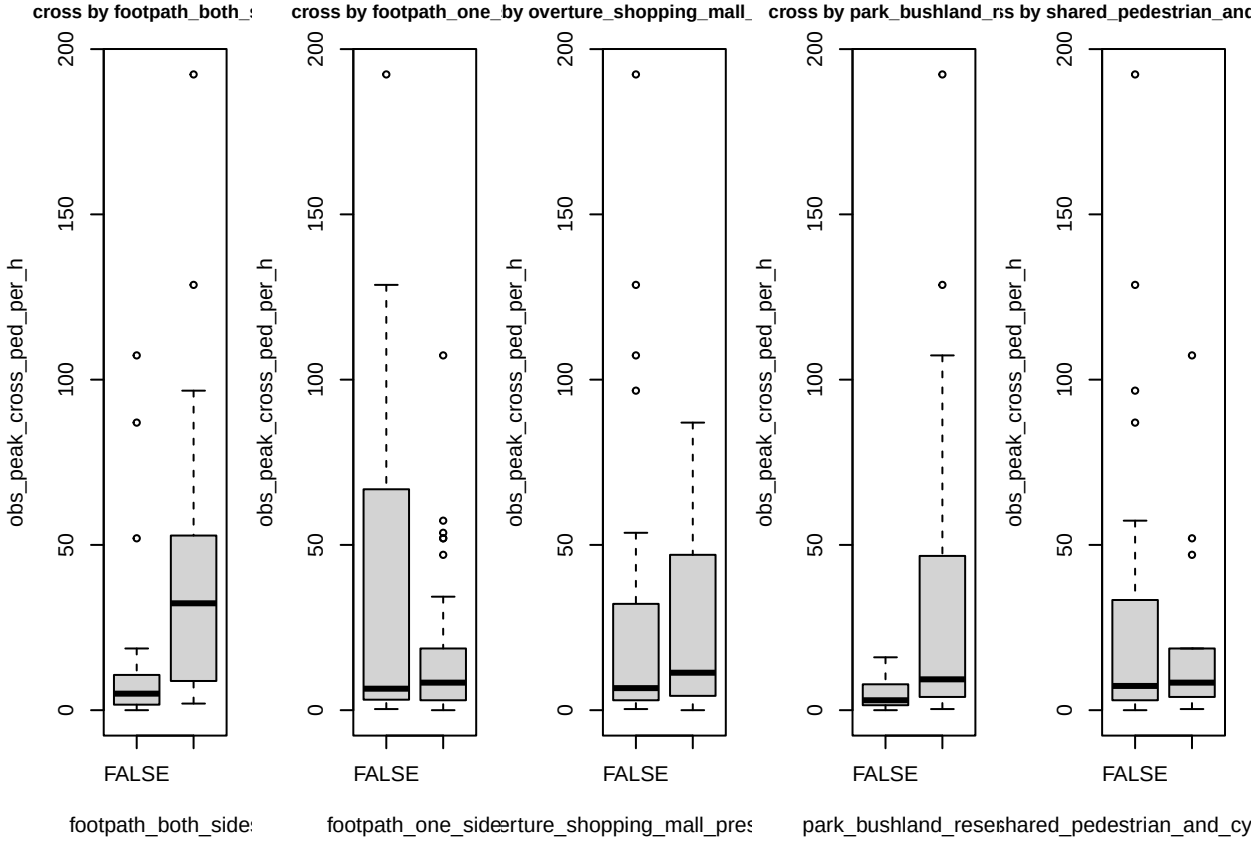
Table 1: TRUE/FALSE counts for logical variables (n=49)

	variable	FALSE_n	TRUE_n
overture_shopping_mall_present_500m	overture_shopping_mall_present_500m	40	9
footpath_both_sides	footpath_both_sides	30	19
footpath_one_side	footpath_one_side	16	33
fencing_barriers	fencing_barriers	3	46
railway_station	railway_station	47	2
bus_stops	bus_stops	47	2
overpass_bridge	overpass_bridge	48	1
park_bushland_reserve	park_bushland_reserve	8	41
shared_pedestrian_and_cycle_path	shared_pedestrian_and_cycle_path	36	13

Four variables have fewer than 5 cases in one group — `fencing_barriers`, `railway_station`, `bus_stops`, and `overpass_bridge` — and are excluded from further analysis, as the small group size makes comparisons unreliable.

The remaining variables are examined with boxplots.


along by footpath_both_
along by footpath_one_by overture_shopping_mall_
along by park_bushland_ng by shared_pedestrian_and
obs_peak_along_ped_per_h
0
50
100
150
200
250
300
350
FALSE
footpath_both_side
footpath_one_side
erture_shopping_mall_pres
park_bushland_reser
shared_pedestrian_and_cy


cross by footpath_both_
cross by footpath_one_by overture_shopping_mall_
cross by park_bushland_rs by shared_pedestrian_and
obs_peak_cross_ped_per_h
0
50
100
150
200
FALSE
footpath_both_side
footpath_one_side
erture_shopping_mall_pres
park_bushland_reser
shared_pedestrian_and_cy

Findings:

`footpath_both_sides` shows a clear pattern: segments with footpaths on both sides (`TRUE`) have a higher median and wider spread for both response variables, compared to segments without (`FALSE`).

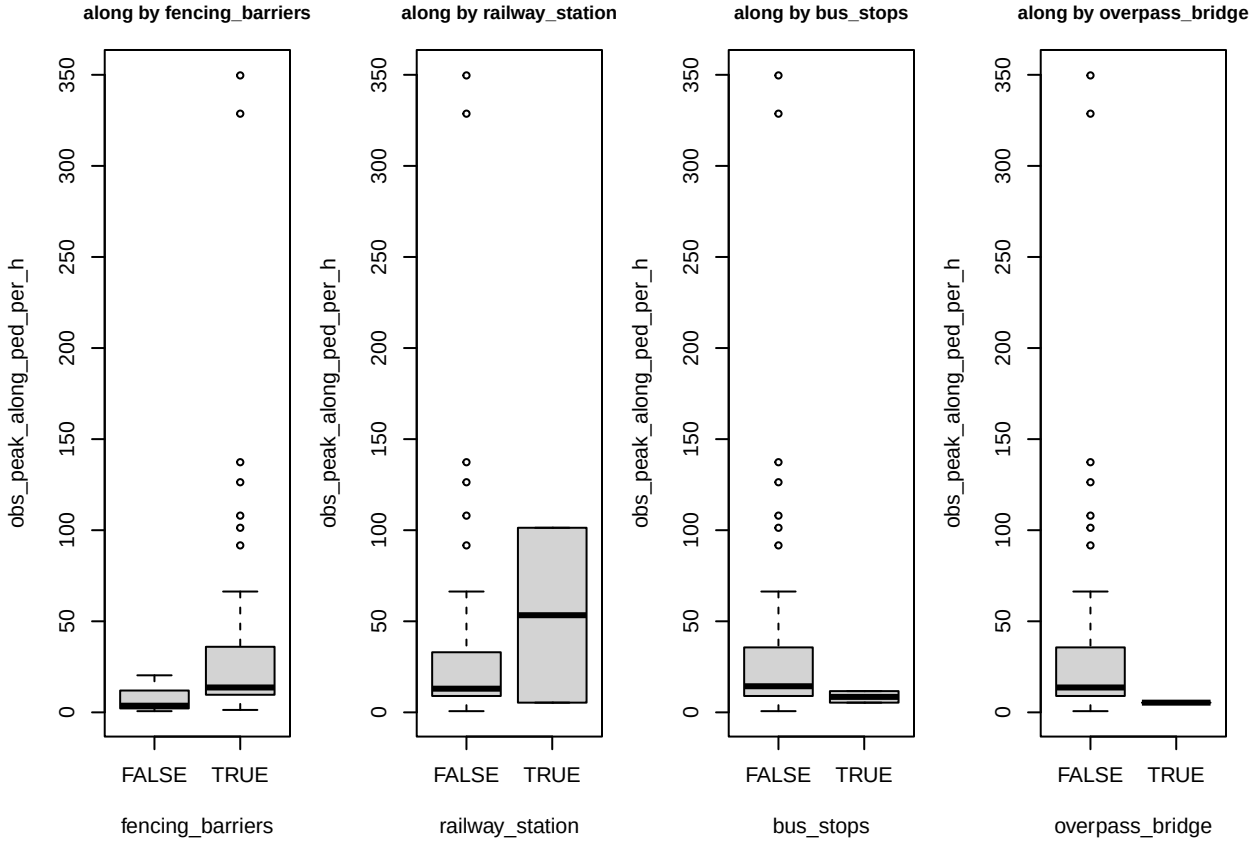
`footpath_one_side` does not show a clear increase in pedestrian counts when `TRUE`; medians and spread are similar between the two groups for both response variables.

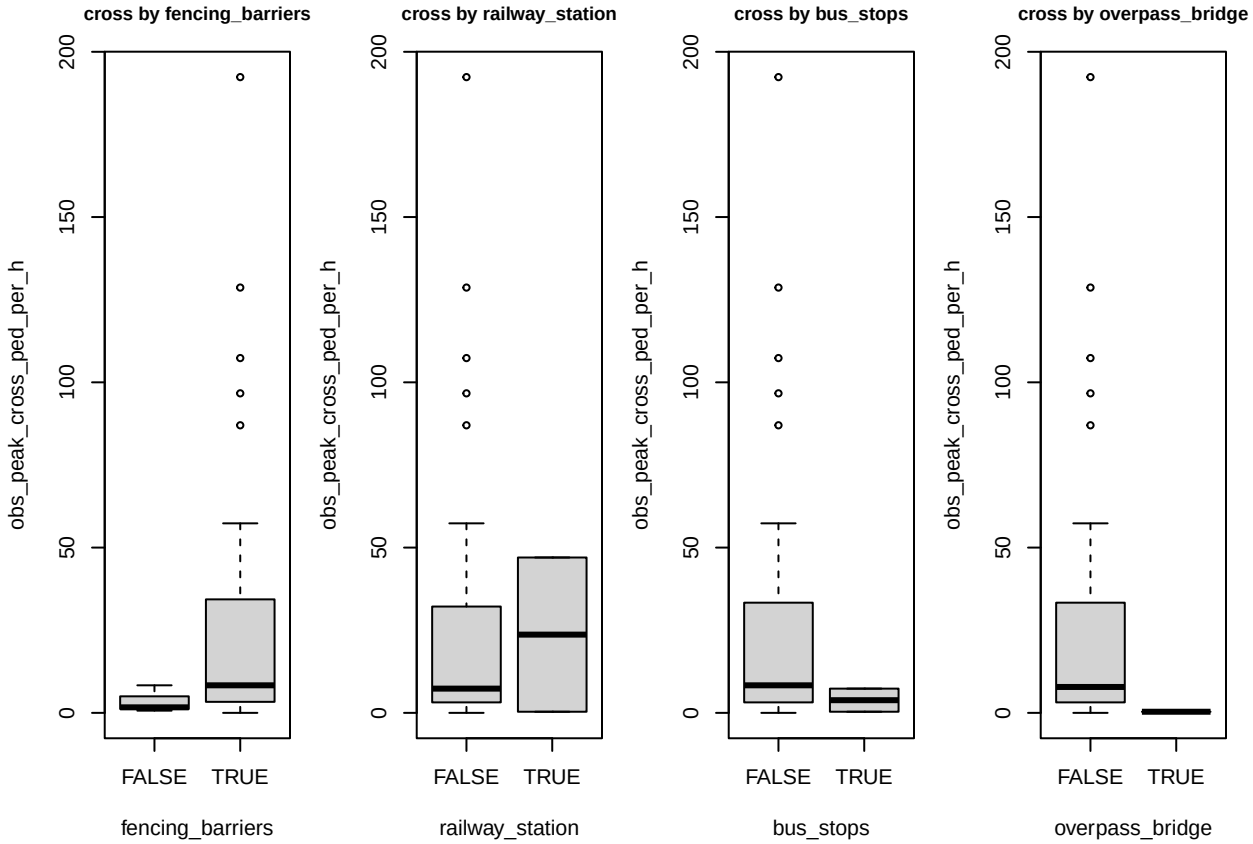
`overture_shopping_mall_present_500m` shows only a modest difference between groups: locations where a shopping mall is present within 500m (`TRUE`) have a slightly higher median than locations where it is absent, but the spread is broadly similar for both target variables.

`park_bushland_reserve` shows a similar pattern to `footpath_both_sides`: locations where this is `TRUE` have a higher median and wider spread than locations where it is `FALSE`, for both target variables.

`shared_pedestrian_and_cycle_path` shows little difference between the two groups, with overlapping boxes and similar medians for both response variables.

As a reference, boxplots for the four excluded variables are shown below. Although the small group size (fewer than 5 cases in one group) makes formal comparisons unreliable, the plots are included to check for any obvious trend before ruling these variables out entirely.





Findings:

`fencing_barriers` and `railway_station` show a trend toward higher pedestrian counts when `TRUE`, for both response variables. However, the `FALSE` group for `fencing_barriers` (n = 3) and the `TRUE` group for `railway_station` (n = 2) are extremely small, so these apparent trends should be treated with caution rather than as reliable evidence of an effect.

`bus_stops` and `overpass_bridge` show the opposite direction, with lower pedestrian counts when `TRUE`, but the `TRUE` groups are similarly small (n = 2 and n = 1 respectively), so no reliable conclusion can be drawn.

Given the small group sizes across all four variables, none are considered suitable for further statistical analysis at this stage.

Character-Type Categorical Variables

The level distribution of each character-type categorical variable was checked.

Table 2: Level counts and percentages for character categorical variables

variable	level	n	pct
abs_mb_category_2021	Commercial	10	20.4
abs_mb_category_2021	Education	3	6.1
abs_mb_category_2021	Industrial	3	6.1
abs_mb_category_2021	Parkland	4	8.2
abs_mb_category_2021	Primary Production	1	2.0
abs_mb_category_2021	Residential	26	53.1
abs_mb_category_2021	Transport	2	4.1
pta_nearest_active_stop_type	School	10	20.4

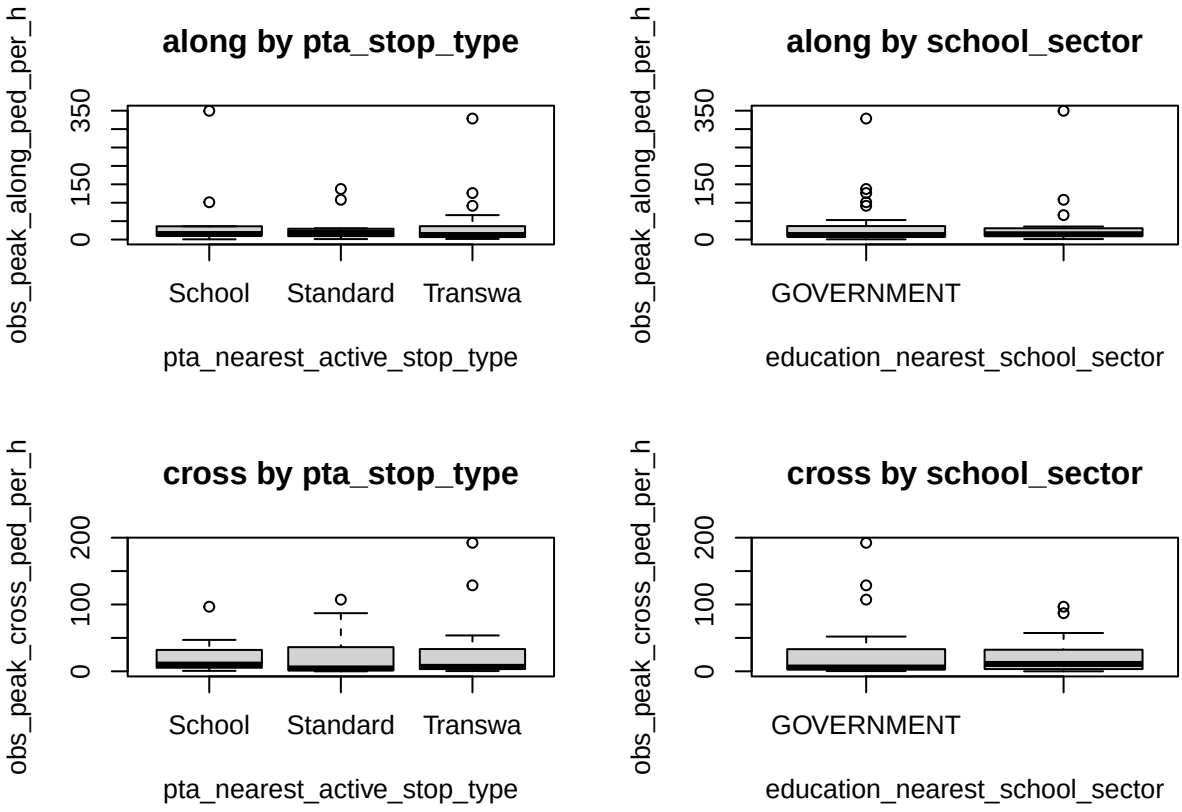
variable	level	n	pct
pta_nearest_active_stop_type	Standard	11	22.4
pta_nearest_active_stop_type	Transwa	28	57.1
health_nearest_establishment_category	Acute Hospital	44	89.8
health_nearest_establishment_category	Nursing Post	5	10.2
health_nearest_establishment_type	Private	4	8.2
health_nearest_establishment_type	Public	45	91.8
education_nearest_school_sector	GOVERNMENT	31	63.3
education_nearest_school_sector	NON-GOVERNMENT	18	36.7
education_nearest_school_low_year	KIN	35	71.4
education_nearest_school_low_year	PKG	8	16.3
education_nearest_school_low_year	Y07	4	8.2
education_nearest_school_low_year	Y10	1	2.0
education_nearest_school_low_year	Y11	1	2.0
education_nearest_school_high_year	Y06	32	65.3
education_nearest_school_high_year	Y10	4	8.2
education_nearest_school_high_year	Y12	13	26.5

Findings:

Two variables have a reasonable balance across all levels: `pta_nearest_active_stop_type` (School=10, Standard=11, Transwa=28) and `education_nearest_school_sector` (GOVERNMENT=31, NON-GOVERNMENT=18).

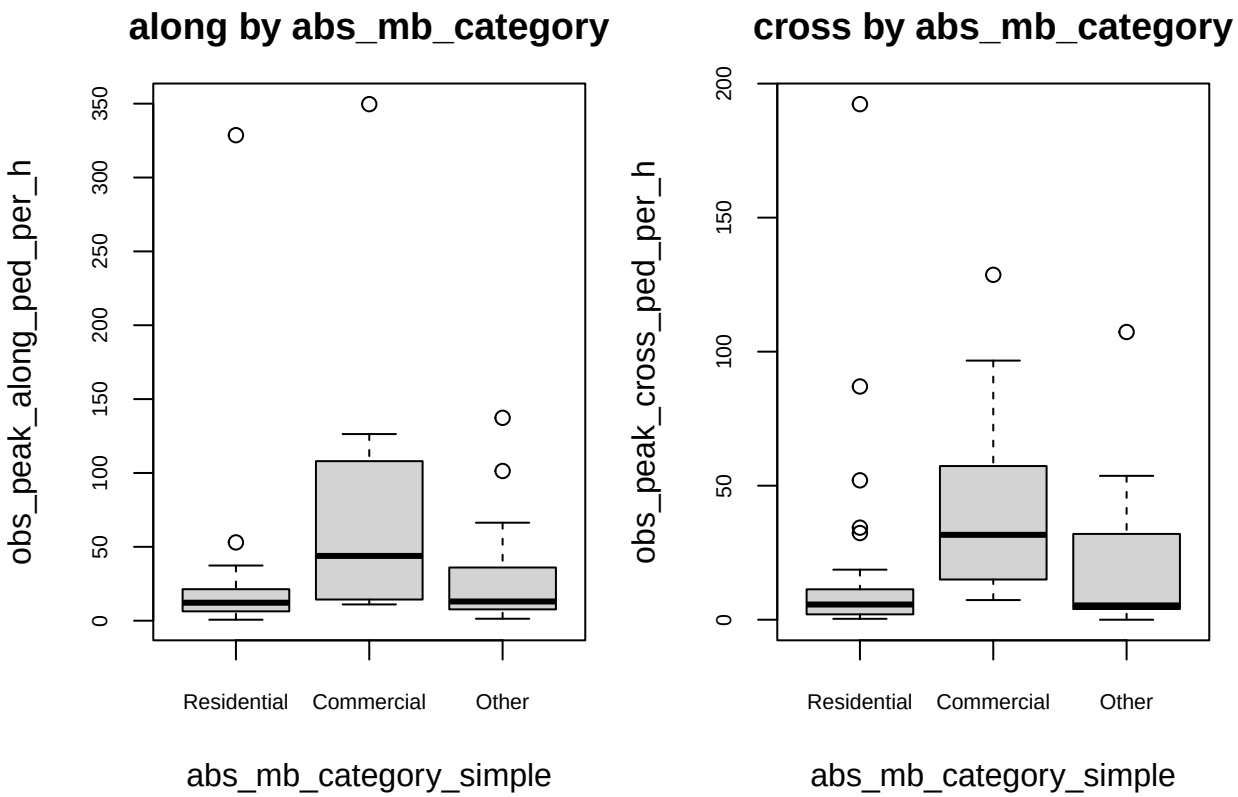
`abs_mb_category_2021` is dominated by Residential (53.1%) and Commercial (20.4%), with the remaining five levels each below 10% of the sample.

The other four variables have at least one level with fewer than 5 observations — `health_nearest_establishment_category`, `health_nearest_establishment_type`, `education_nearest_school_low_year`, and `education_nearest_school_high_ye` — and are excluded from further analysis for the same reason as above.



Neither `pta_nearest_active_stop_type` nor `education_nearest_school_sector` shows a meaningful difference in pedestrian counts across groups, based on these plots.

`abs_mb_category_2021` is simplified into three levels — `Residential`, `Commercial`, and `Other` (combining Education, Industrial, Parkland, Primary Production, and Transport) — since the smaller levels each account for less than 10% of the sample.



The Commercial group shows the highest median and widest spread, followed by Other, with Residential showing the lowest and narrowest distribution, for both target variables. This is consistent with the earlier finding that `commercial_buildings` was the strongest numeric predictor.

Conclusion

`commercial_buildings` and related retail/commercial density measures (`overture_other_shopping_count_500m`, `overture_grocery_convenience_pharmacy_count_500m`, `parking_areas`, `overture_food_drink_count_500m`) are the strongest and most consistent numeric predictors of pedestrian counts, but they are highly correlated with each other and require a multicollinearity check before joint use in a model.

Among categorical variables, `footpath_both_sides`, `overture_shopping_mall_present_500m`, `park_bushland_reserve`, and land-use category (`abs_mb_category_simple`) show a clear relationship with pedestrian counts, all pointing toward higher counts in commercially developed, well-footpathed areas. `footpath_one_side`, `shared_pedestrian_and_cycle_path`, `pta_nearest_active_stop_type`, and `education_nearest_school_sector` show no meaningful relationship and are unlikely to add value to the model.

Recommended next step: examine multicollinearity among the commercial/retail density variables (e.g. via VIF or correlation-based selection) before fitting a regression model, and confirm whether `footpath_both_sides` and `abs_mb_category_simple` add explanatory power independent of `commercial_buildings`.